THE NATIONAL UNIVERSITY OF LESOTHO

COURSE OUTLINE			
Course Title	Analog Communications		
Course Code	EE3408	**No. Of Credits**	11
Department	Physics and Electronics	**Faculty**	Science and Technology
Pre-requisites Course Code		**Co- requisites Course Code**	
Course Coordinator(s)	M. Makhele		
Email	makhelem@gmail.com	**Telephone No.**	52213528 Ext. 3528
Other Course Teachers(s) Tutor(s)	R. Sehobai Email: reabetsoesehobai.rs@gmail.com		
Learning Hours	Total: 110 Learning Hours Lecture (24hrs), Laboratory (24hrs), Independent Study (48 hours), Tests and Exam (14 hours)		
Student Quota			
Course Type	☐ Core Course ☐ Compulsory ☐ Elective ☐ Other.....		
Offer in Academic Year	☐ 1st Semester ☐ 2nd Semester ☐ Year Long		

COURSE DESCRIPTION

This course provides a comprehensive coverage of the techniques (amplitude and frequency modulation and their variations) and circuits used in traditional analogue communication systems. The course develops the engineering mathematics to describe the physical transmission of information bearing signals over point-to-point links, taking into account channel characteristics and presence of noise and distortion.

COURSE OBJECTIVES

The course objectives are:

1. to familiarize students with the concepts of Fourier series, functions of filters, modulators, limiters, mixers, detectors in AM, FM, PM, SSB and PLL circuits
2. to equip and impart knowledge on the principles underlying modulation and demodulation of analogue signals with associated system design issues, taking into account power and bandwidth constraints, as well as performance in the presence of additive noise
3. to provide comprehensive, hands-on instruction in the terminology, principles, and applications of the analogue communication circuits

GUIDELINES ON GRADING AND STANDARDS OF ASSESSMENT

We follow a structured framework to ensure fairness, transparency, and academic integrity.

- Clear communication and assessment criteria and methods.
- Consistency in assessment outcomes.
- Equal opportunity for everyone to demonstrate their knowledge and skills.
- There are ongoing assessments (coursework) aimed at providing feedback to improve student learning.
- Final exam at the end of a course to assess students learning against the objective.
- Timely and constructive feedback to help students understand their performance and improve

COURSE CONTENT

Course topics include:

(a) Introduction to general communication theory

- the need to transmit information
- Analogue and digital signals
- the communication channel
- the problems of noise and interference

(b) Representation and analysis of signals

- Time and frequency domain representation of signals
- Simple and complex signals
- Periodic and aperiodic signals
- Fourier analysis of periodic signals and Fourier transforms of aperiodic signals
- Spectrum of signals
- Filters, band-limiting, band-limited functions and their significance in telecommunications, sampling

(c) Analogue Modulation (amplitude and angle modulation schemes)

- Amplitude modulation: modulation index, power in AM signals, Bandwidth in DSB-C, DSB-SC, SSB and Vestigial sideband (VSB) modulations
- Angle modulation: Frequency Modulation (FM), Phase Modulation (PM), FM and PM relationships, Frequency deviation, modulation index, bandwidth in FM, Bessel function

(d) **Modulators and Demodulators** (Generation and detection of amplitude and angle modulation schemes)

- Balanced modulators
- AM generations and detections of DSB-C, DSB-SC, SSB and VSB
- FM generation and detection

(e) **Multiplexing** (Frequency and time division multiplexing)

- Frequency division multiplexing (FDM), FDM systems
- Time division multiplexing (TDM), Statistical TDM, TDM systems
- Quadrature Multiplexing

COURSE LEARNING OUTCOMES

Course Learning Outcomes	**Aligned Programme Learning outcomes**
Hard Skills (Subject specific) After the course completion students would be able to: H1 evaluate analogue communication systems including the elements of an analogue transmission system H2 understand the limitation of communication channels H3 Model and implement different analogue modulation schemes H4 understand the effect of noise and interference in the performance of analogue communication systems **Soft Skills** (Generic)	

After the course completion students would be able to: S1 Undertake independent study (Life-long Learning) S2 Access learning material online (Digital Literacy) S3 Work together in small groups (team work) S4 Produce technical reports on assigned tasks (writing skills) S5 Communicate concepts analogue communication systems (Communication skills)	

TEACHING AND LEARNING ACTIVITIES

`Student will acquire the skills by: ➢ Doing some independent study on the material provided in the learning management tool (Tool) ➢ Attending lectures and engaging in interactive class discussion ➢ Working in groups to tackle some of the assigned tasks/assignments ➢ Simulating and modelling some of the communication systems ➢ Building, analysing and evaluating some of the communication systems circuits ➢ Writing reports on the labs undertaken ➢ Doing some presentations on some of the tasks given. ➢ Undertaking laboratory and tutorial work	

COURSE ASSESSMENT METHODS

Assessment Method	Description	Weight	Aligned Course Learning outcomes
COURSEWORK			
➢ Assignments	Assignments in line with the topics being covered will be given. This includes quizzes and presentations.	5%	
➢ Tests	A minimum of three invigilated tests will be given covering two topics, taking into consideration the size of each topic.	25%	
➢ Labs	A minimum of three laboratories based on the topics covered will be conducted. Students will be assessed based on their lab-reports taking into consideration the student's attendance.	20%	
EXAMINATION ➢ Exam	An end of semester examination will entail a combination of short and long	50%	

FINAL MARK	responses and will be three hours in length. The final mark will be composed of 50% Coursework and 50% Final examination mark		

ESSENTIAL READINGS: (Journals, textbooks, website addresses, etc)

Textbooks:

1. Stremler, Ferrel G., *Introduction to Communication Systems*, 3rd edition, Addison-Wesley Publishing Company, 1990.
2. B.P. Lathi, *Modern Digital and Analog Communication Systems, 3rd edition, McGrqw Hill, 2000*
3. John G. Proakis & Masoud Salehi, *Communication Systems Engineering*, 2nd edition, Prentice Hall, 2002
4. Wayne Tomasi, *Electronic Communications Systems: Fundamentals through Advanced,* 3rd Editon, Prentice Hall, 1998
5. Simon Haykin, Michael Moher, *An Introduction to Digital and analog Communications*, John Wiley and Sons, 1989.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

COURSE POLICY (including plagiarism, academic honesty, attendance, etc)

Unless otherwise stated, the normal course administration policies and rules of the Fuculty of Science and Technology apply. For the policy, see http://www.nul.ls

The University Regulations on academic misconduct will be strictly enforced. Please refer to http://www.nul.ls/student/plagiarism for further details. In addition, students should familiarise themselves with http://www.nul.lib.ls

General health and safety information can be obtained from http://www,nul.ls/health-and-safety

ADDITIONAL COURSE INFORMATION

The Course Coordinator is available for consultation at times that is displayed on the coordinator's office notice. Queries may also be emailed to the Course Coordinator directly at makhelem@gmail.com

ACADEMIC COUNCIL APPROVAL

Signature: Date: